

Density functional studies on a novel double-shell fullerene C₂₀@C₆₀

Fengyi Liu^{a,b}, Lingpeng Meng^b, Shijun Zheng^{a,b,*}

^aOpen Laboratory of Computational Quantum Chemistry, Hebei Normal University, Hongqi Street, Shijiazhuang, Hebei Province 050091, China

^bScience Faculty, Beijing University of Chemical Technology, Beijing 100029, China

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Abstract

Geometrical and electronic structures of the smallest potential double-shell fullerenes C₂₀@C₆₀ were studied by using Hartree–Fock and density functional theory (DFT) B3LYP methods. Two distinctly different highly endothermic but stable isomers are located; the fusiform one with *D*_{5d} symmetry is 848.22 kcal mol^{−1} lower in energy than the spherical (*I*_h) one. Their frontier orbital, electrostatic potential as well as the electron affinity (EA) and the ionization potential (IP) were presented to give a comprehensive description of this novel endohedral system. The inter-shell interactions between C₂₀ layer and C₆₀ layer in C₂₀@C₆₀ are proved to be ‘through-bond’ and ‘through-space’ interactions for the fusiform isomer and spherical one, respectively.

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1. Introduction

Since the discovery [1] and macroscopic scale synthesis [2] of C₆₀, carbon clusters have been extensively researched for the purpose of producing new materials with promising physical and chemical properties. Single-wall fullerenes from C₂₀ up to C₉₆ have been detected by anion photoelectron spectroscopy and mass spectroscopy [3] or synthesized in macroscopic quantities [4] and numerous theoretical calculations have been performed to characterize them [5–8]. Recently, an interesting and novel variety of carbon clusters, multi-shell fullerenes including C₆₀@C₂₄₀, C₂₄₀@C₅₆₀ and C₈₀@C₂₄₀@C₅₆₀ were observed in the laser pyrolysis carbon black [9,10] and synthesized by laser vaporization method with a yield of mixed multi-shell fullerenes up to 90% [11]. If it is possible to isolate multi-shell fullerene crystals, it is interesting to compare their properties with that of other endohedrally doped fullerenes

[12–15]. Unfortunately, it is extreme difficulty to produce macroscopic quantities and isolate pure samples; therefore, theoretical studies have been playing an important role in investigating the structural and electronic properties and predicting the potential characteristics of multi-shell fullerenes as well as explaining their formation mechanisms.

C₂₀@C₆₀, which was first suggested by L. Türker [16], should constitute one of the simplest bucky onion-like fullerenes. In fullerene family, C₂₀ is the smallest possible fullerene without hexagonal rings. It contains just 12 pentagons, in a highly strained structure. The stable isomers of C₂₀ clusters consist of bowl, ring, and fullerene (cage) structures [17], each of them can be experimentally produced under suitable reaction conditions [18]. Wherein, fullerene (*I*_h) is the most stable conformer [17]. The structure of C₆₀, which has 20 hexagonal and 12 pentagonal rings, has been well-characterized both experimentally and theoretically [1,17]. Since the radius of C₆₀ is approximately 3.5 Å, the C₂₀ fullerene with a radius about 2.0 Å could be placed inside the sphere. Hence, in the present study, the structural and electronic properties of simplest multi-shell C₂₀@C₆₀ fullerenes were studied by using Hartree–Fock and density functional theory, and we hope our results may shed some light on the understanding of such fullerene systems.

* Corresponding author. Address: Open Laboratory of Computational Quantum Chemistry, Hebei Normal University, Hongqi Street, Shijiazhuang, Hebei Province 050091, China. Tel.: +86 311 6263381; fax: +86 310 6263269.

E-mail address: sjzheng@mail.hebtu.edu.cn (S.J. Zheng).

2. Computational methods

Geometry optimizations of $C_{20}@C_{60}$ were performed at the Hartree–Fock and DFT/B3LYP level of theory with the less sophisticated STO-3G* and 3-21G* basis sets due to the large amount of atoms. Harmonic vibrational frequencies were calculated at Hartree–Fock level to ensure true minima were located. The total energies, orbital energies and molecular electrostatic potential (MEP) were obtained from B3LYP/3-21G* calculation with SCF convergence criteria 1×10^{-8} . To get the electron affinity (EA) and the ionization potential (IP) of different isomers of $C_{20}@C_{60}$, total energies for $C_{20}@C_{60}^+$ and $C_{20}@C_{60}^-$ were also calculated at the same level. All calculations were carried out with the GAUSSIAN 98 program package [19].

3. Results and discussions

Three models, HF/STO-3G*, HF/3-21G* and B3LYP/3-21G*, were employed in geometry optimizations to obtain results at different theory levels as well as to minimize the computational cost. The single-wall fullerenes C_{20} and C_{60} were firstly optimized, and then the $C_{20}@C_{60}$ was constructed by putting C_{20} in the position of the spherical center of C_{60} along different directions. In fact, due to the inter-shell distances in $C_{20}@C_{60}$ are quite short, the carbon atoms in C_{20} layer are strongly confined by the outer-shell, as results, only two highly symmetric isomers, Isomer I (D_{5d}) and II (I_h), were located in geometry optimization at Hartree–Fock and B3LYP level. Their optimized geometries are summarized in Fig. 1 and Table 1.

From the viewpoint of double-shell fullerenes, both isomers I and II of $C_{20}@C_{60}$ show shorter inter-shell distances than the observed average inter-layer separation 3.4 Å in $C_{60}@C_{240}$ [10]. The B3LYP/3-21G* calculation gave average radii 1.988 Å for C_{20} shell and 3.814 Å for C_{60} shell in isomer I, the corresponding values in isomer II are 1.917 Å for C_{20} and 3.863 Å for C_{60} layer, while the spherical radius for free C_{20} is 2.046 Å and free C_{60} is 3.555 Å. Obviously, in both conformers of $C_{20}@C_{60}$, the inner-shells are compressed while the outer-shells are expanded due to the inter-shell repulsion caused by these over-crowded structures. It is also interesting to see from Fig. 1 that Isomer I and II have distinctly different structures: Isomer I shows an ‘elongate’ fusiform geometry along the z -axis (see Fig. 1) while Isomer II keeps the spherical structure. This structural difference between them should be illuminated by analyzing the inter-shell interactions between C_{20} and C_{60} layer in both conformers. In isomer I with D_{5d} symmetry, each atom (e.g. C(1) atom, etc.) in the top pentagon of C_{20} directs towards a corresponding carbon (e.g. C(26), etc.) in C_{60} layer, the distance between them is only 1.535 Å at B3LYP/3-21G* level and it is evidently to see the existence of inter-shell C–C bond. The formations of chemical bonds change the hybrid states of C(1) and other equivalent atoms in C_{20} layer as well as corresponding atoms (e.g. C(26)) in C_{60} layer from sp^2 to sp^3 , which further weaken the tensility and conjugation in the hexagons of C_{60} layer. As results, the C(26) and other equivalent atoms are sunken-in and the topmost pentagon formed by C(21)–C(25) is pushed away from its original position in free C_{60} ; the isomer I itself shows an elongated fusiform symmetry. In the symmetry-holding isomer II, each pentagon in C_{60} layer is central projected to a pentagon of inner-shell in adverse orientation

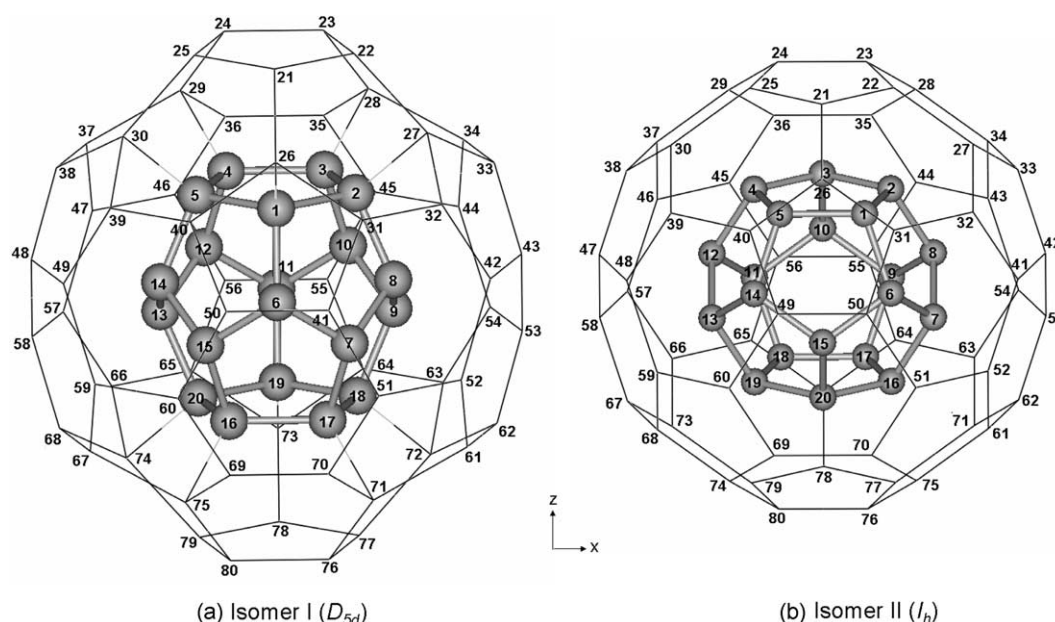


Fig. 1. Optimized geometries of $C_{20}@C_{60}$ isomers. The inner C_{20} shell is plotted in ball and stick and outer C_{60} shell in wire-frame.

Table 1
Optimized bond lengths and inter-shell distances (Å) of C₂₀@C₆₀ at Hartree–Fock and B3LYP level

Isomer	Parameter ^a	HF/ STO-3G*	HF/ 3-21G*	B3LYP/ 3-21G*
I (<i>D</i> _{5d})	C(1)–C(2)	1.474	1.464	1.479
	C(1)–C(6)	1.441	1.416	1.429
	C(6)–C(7)	1.421	1.345	1.362
	C(1)–C(26)	1.525	1.520	1.535
	C(21)–C(22) (h–p) ^b	1.455	1.459	1.482
	C(21)–C(26) (h–h)	1.618	1.634	1.626
	C(26)–C(31) (h–p)	1.612	1.613	1.623
	C(31)–C(32) (h–h)	1.466	1.477	1.481
	C(31)–C(41) (h–p)	1.464	1.443	1.474
	C(41)–C(50) (h–p)	1.532	1.530	1.540
	p(inner)–p(outer) ^c	2.185	2.171	2.172
	C(inner)–C=C(outer)	1.659	1.807	1.813
II (<i>I</i> _h)	C(1)–C(2)	1.417	1.353	1.368
	C(21)–C(22) (h–p)	1.493	1.488	1.510
	C(21)–C(26) (h–h)	1.637	1.646	1.645
	p(inner)–p(outer)	2.033	2.107	2.119
	C(inner)–h(outer)	1.508	1.598	1.609

^a Other geometric parameters are omitted according to molecular symmetry.

^b Position of chemical bond: h–p denotes the common border between a hexagon and pentagon and p–p means the bond belongs to two hexagon.

^c Inter-shell distance: p(inner)–p(outer) denotes the distance between pentagons of C₂₀ layer and the aspectant one of C₆₀ layer; C(inner)–C=C(outer), the distance between the carbon atom of C₂₀ shell and the center of C=C bond in outer C₆₀ shell; C(inner)–p(outer), the distance between the inner-shell carbon and center of aspectant outer-shell hexagon.

(see Fig. 1), the repulsion between them enlarges the radius of outer-shell and compress the inner C₂₀ shell. The B3LYP/3-21G* results show a inter-shell spacing about 1.95 Å, which is much larger than the radius difference of 1.51 Å between free C₆₀ and C₂₀ fullerenes calculated at the same level.

The total/relative energies and the atomization energies as well as the frontier orbital energies of C₂₀@C₆₀ are shown

in Table 2. The relative energies from DFT/B3LYP calculation show Isomer I with *D*_{5d} symmetry locates 848.22 kcal mol^{−1}, which is lower in energy than the spherical isomer II (*I*_h). In fact, both isomer I and II are highly endothermic, their atomization energies per carbon, −182.07 and −171.46 kcal mol^{−1}, respectively, are evidently lower than the average value −200.23 kcal mol^{−1} of C₆₀; meanwhile, the former is slightly larger than −182.02 kcal mol^{−1} of C₂₀ but the latter is less stable than C₂₀. Hence, their relative stabilities should be in the order of C₆₀ > C₂₀@C₆₀ (*D*_{5d}) > C₂₀ > C₂₀@C₆₀ (*I*_h). Moreover, the HOMO–LUMO gap is traditionally associated with chemical stability against electronic excitation, with larger gap corresponding to greater stability. The HOMO–LUMO gaps of Kohn–Sham orbital calculated at B3LYP/3-21G*//B3LYP/3-21G* level for isomer I and II are 1.139 and 0.885 eV, show evident reductions compared with the gap value 1.666 eV of C₆₀ [20], also suggestive of the reduced stabilities from C₆₀, C₂₀@C₆₀ (*D*_{5d}) to C₂₀@C₆₀ (*I*_h).

Fig. 2 shows the three-dimensional contour diagrams of HOMO and LUMO in isomer I and II. It is seen that both the double degenerated HOMO (*e*_{1u}) and the LUMO (*a*_{1g}) orbital of isomer I have approximate equivalent contributions from inner-shell and outer-shell; while in isomer II, the main characters of HOMO (*t*_{1g}) orbital are from outer C₆₀ layer and the dominant coefficients of LUMO (*h*_g) are from inner C₂₀ layer. It reflects the inter-shell interactions in these conformers are different. These interactions will be discussed below on the bases of atomic charge distribution and molecular electrostatic potential.

The ionization potential (IP) and electron affinities (EA), which are directly relative to the possibilities of donating electron or accepting one electron, respectively, are also seen in Table 2. Here we give the B3LYP/3-21G*//B3LYP/3-21G* vertical values of IP and EA, the former corresponds

Table 2
Some energies of C₂₀@C₆₀ at B3LYP/3-21G* level based on Hartree–Fock and B3LYP optimized geometries

Isomer	Energy	B3LYP//HF ^a	B3LYP//HF ^b	B3LYP//B3LYP
I (<i>D</i> _{5d})	Total (Hartree)	−3028.9807460	−3029.0228152	−3029.0460924
	Δ <i>E</i> (kcal mol ^{−1})	0.00	0.00	0.00
	Atomization (kcal mol ^{−1})	−14524.51	−14550.91	−14565.51
	HOMO (eV)	5.359	5.495	5.459
	LUMO (eV)	4.237	4.248	4.319
	Δ <i>E</i> _{HOMO–LUMO} (eV)	1.112	1.247	1.139
	IP (eV)			6.688
	EA (eV)			3.234
II (<i>I</i> _h)	Total (Hartree)	−3027.5256091	−3027.6680044	−3027.6943619
	Δ <i>E</i> (kcal mol ^{−1})	913.11	850.16	848.22
	Atomization (kcal mol ^{−1})	−13611.40	−13700.75	−13717.29
	HOMO (eV)	5.920	6.154	5.882
	LUMO (eV)	4.941	5.218	4.997
	Δ <i>E</i> _{HOMO–LUMO} (eV)	0.979	0.936	0.885
	IP (eV)			7.356
	EA (eV)			3.931

^a B3LYP/3-21G* energies based on HF/STO-3G*.

^b HF/STO-3G* optimized geometries.

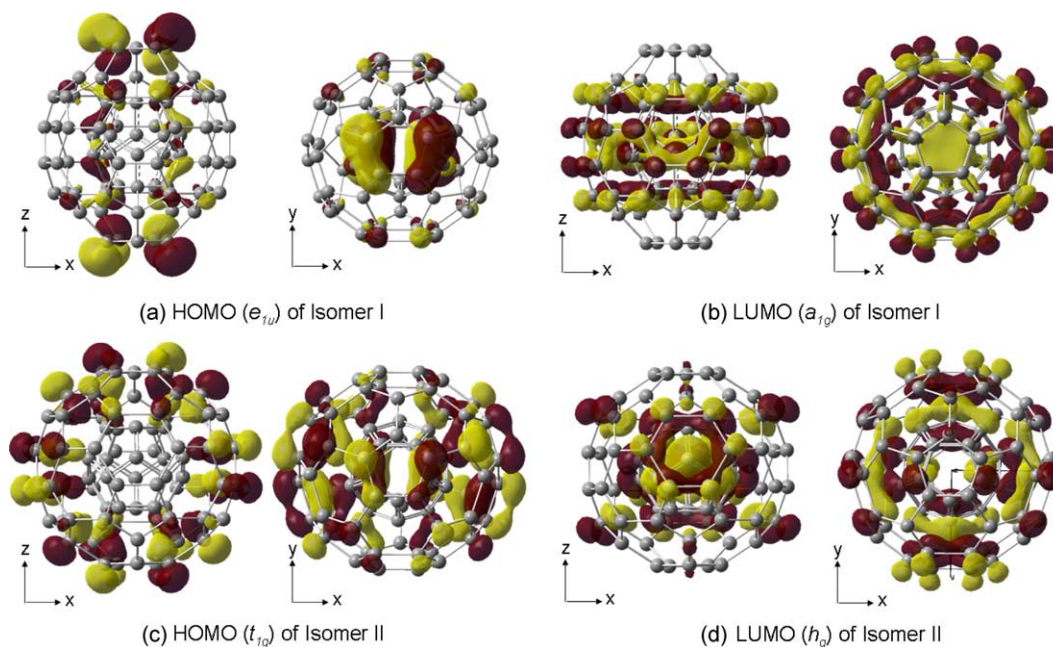


Fig. 2. Diagrams of HOMO and LUMO orbitals for C₂₀@C₆₀ isomers with contour value = 0.025.

to the energy difference between the cationic state in the geometry of neutral state and the optimized neutral state, whereas the latter refers to the energy difference between the optimized neutral state and anionic state in the geometry of neutral state. The VIP values are 6.688 and 7.356 eV for isomer I and II, respectively, slightly lower than 7.859 eV for C₆₀ calculated at the same level. While the VEA values of these two isomers are 3.234 and 3.931 eV, respectively, evidently larger than 1.866 eV of C₆₀. These results demonstrate the stability of double-shell C₂₀@C₆₀ and also show it has similar possibility of both accepting and donating electrons with the single-shell C₆₀ fullerene.

Table 3 lists the Mulliken atomic charges of C₂₀@C₆₀ calculated at different levels. The atomic changes of carbon atoms in free *I_h* symmetrical C₆₀ and C₂₀ fullerenes are all zero; however, the C₂₀@C₆₀ shows interesting atomic charge populations due to the structural distortions. The atoms of inner C₂₀ layers in both isomers show positive charges, but those of outer-shells are quite different. In isomer I, the atoms of C₆₀ layer could be divided into four

kinds in which the atoms belongs to the topmost pentagon show negative charges −0.134 (e.g. C(21), etc. B3LYP/3-21G* value). It may imply the existence of ‘through-bond’ interactions in which the inner-C₂₀ is electron donor and the two topmost/downmost pentagons in C₆₀ layer are electron acceptor, their neighboring atoms, i.e. C(26) and other equivalent atoms, connecting both to the outer-layer

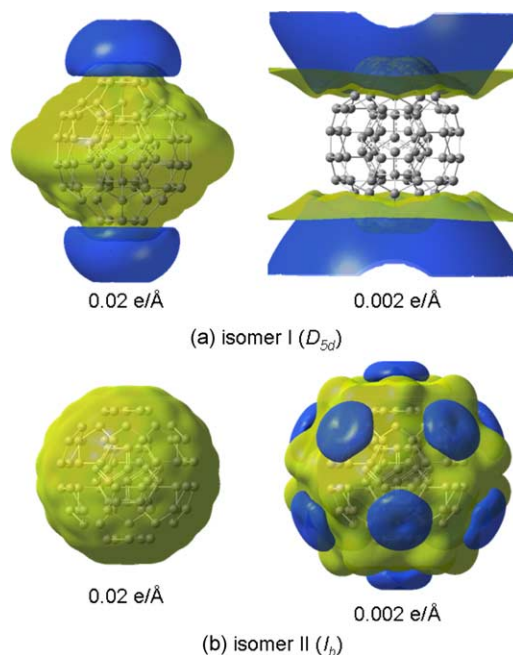


Fig. 3. Three-dimensional molecular electrostatic potential of C₂₀@C₆₀ isomers. The positive value is in yellow and negative value in blue. (For interpretation of the reference to colour in this legend, the reader is referred to the web version of this article.)

Table 3

Mulliken atomic charges of C₂₀@C₆₀ at Hartree–Fock and B3LYP level

Isomer	Atom ^a	HF/STO-3G*	HF/3-21G*	B3LYP/3-21G*
I (<i>D</i> _{5d})	C(1)	0.071	0.023	0.028
	C(6)	−0.047	0.020	0.023
	C(21)	−0.008	−0.015	−0.013
	C(26)	−0.152	−0.136	−0.134
	C(31)	0.079	0.076	0.072
	C(41)	−0.011	−0.022	−0.023
II (<i>I</i> _h)	C(1)	0.065	0.078	0.083
	C(21)	−0.022	−0.026	−0.028

^a Atomic charges of other atoms are omitted according to molecular symmetry.

and to the inner layer, act as bridged-atoms in the inter-shell interaction. In isomer II, the atomic charges of C_{60} layer are -0.028 (B3LYP/3-21G* results), denotes another type of ‘through-space’ interactions between the pentagons in C_{20} layer (as electron donors) and C_{60} layer (as electron acceptors). That is, two different interactions could be found in different isomers of $C_{20}@C_{60}$ system.

The inter-shell interactions in $C_{20}@C_{60}$ can also be revealed by analyzing their molecular electrostatic potential (MEP). The MEP has been employed as an informative tool of chemistry to describe physical properties, molecular reactivities, and intermolecular interactions [21,22]. In particular, noncovalent molecular interactions seem likely to depend upon the patterns of the electrostatic potentials over the molecules’ entire surfaces as well as the local maxima and minima. Here we introduce the MEP to describe the inter-shell interaction in $C_{20}@C_{60}$ fullerenes. The three-dimensional MEP surfaces of both conformers are plotted in Fig. 3. Both of them show quite different patterns compared with the spherical potential of C_{20} and C_{60} [16]. In isomer I, the negative potential concentrates over the surface of the upmost/downmost pentagons in C_{60} layer, while in isomer II, small negative potential regions over the pentagons of outer-shell can be seen when contour value decrease to $0.002 \text{ e/\AA}$. In general, the negative electrostatic potentials are usually associated with the lone pairs of the more electronegative atoms and unsaturated aromatic and strained carbon–carbon bonds. It is evident that the negative MEP regions in isomer I and II are caused by the inner-pentagons π conjugations. The through-bond interactions in isomer I as well as the through-space interactions in II increase the negative charges in the certain pentagons to form anion-like $4m$ type π -systems, which are aromatic and stable.

4. Conclusions

In the present study, the smallest potential double-shell fullerenes $C_{20}@C_{60}$ were investigated and two distinctly different highly endothermic but stable isomers are located; the fusiform one with D_{5d} symmetry is $848.22 \text{ kcal mol}^{-1}$ lower in energy than the spherical (I_h) one. Their frontier orbital, electron affinity (EA) and ionization potential (IP) as well as molecular electrostatic potential were presented. All results show the stability of $C_{20}@C_{60}$ and the existence of strong inter-shell interactions between C_{20} and C_{60} layers. These inter-shell interactions, which are proved to be

‘through-bond’ and ‘through-space’ interactions for the fusiform and spherical isomer, respectively, will be addressed in future studies.

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References

- [1] H.W. Kroto, J.R. Heath, S.C. O’Brien, R.F. Curl, R.E. Smalley, *Nature* 318 (1985) 162.
- [2] W. Krätschmer, L.D. Lamb, K. Fostiropoulos, D.R. Huffman, *Nature* 347 (1990) 354.
- [3] H. Kietzmann, R. Rochow, G. Ganteför, W. Eberhardt, K. Vietze, G. Seifert, P.W. Fowler, *Phys. Rev. Lett.* 81 (1998) 5378 and references therein.
- [4] G.E. Scuseria, *Science* 271 (1996) 942 and references therein.
- [5] J.M.L. Martin, *Chem. Phys. Lett.* 255 (1996) 1.
- [6] F. Jensen, H. Koch, *J. Chem. Phys.* 108 (1998) 3213.
- [7] P.R.C. Kent, M.D. Towler, R.J. Needs, G. Rajagopal, *Phys. Rev. B* 62 (2000) 15394.
- [8] P.R. Taylor, E. Bylaska, J.H. Weare, R. Kawai, *Chem. Phys. Lett.* 235 (1995) 558.
- [9] V.Z. Mordkovich, A.G. Umnov, T. Inoshita, M. Endo, *Carbon* 37 (1999) 1855.
- [10] V.Z. Mordkovich, *Chem. Mater.* 12 (2000) 2813.
- [11] V.Z. Mordkovich, Y. Takeuchi, *Chem. Phys. Lett.* 355 (2002) 133.
- [12] J.R. Heath, S.C. O’Brien, Q. Zhang, Y. Liu, R.F. Curl, H.W. Kroto, F.K. Tittel, R.E. Smalley, *J. Am. Chem. Soc.* 107 (1985) 7779.
- [13] R. Tegelmann, N. Krawez, S.H. Lin, I.V. Hertel, E.E. Campbell, *Nature (London)* 382 (1996) 407.
- [14] L.S. Wang, O. Cheshnovsky, R.E. Smalley, J.P. Carpenter, S.J. Hwo, *J. Chem. Phys.* 96 (1992) 4028.
- [15] B. Palpant, Y. Negishi, M. Sanekata, K. Miyajima, S. Nagao, K. Judai, D.M. Rayner, B. Simard, P.A. Hackett, A. Nakajima, K. Kaya, *J. Chem. Phys.* 114 (2001) 8459.
- [16] L. Türker, *J. Mol. Struct. (Theochem)* 545 (2001) 207.
- [17] J. Cioslowski, *Electronic Structure Calculations on Fullerenes and Their Derivatives*, Oxford University Press, New York, 1995.
- [18] H. Prinzbach, A. Weiler, P. Landenberger, E. Wahl, J. Worth, L. Scott, M. Gelmont, D. Olevano, I. BV, *Nature* 407 (2000) 60.
- [19] M.J. Frisch, GAUSSIAN 98, Gaussian, Inc., Pittsburgh, PA, 1998.
- [20] X.Y. Ren, Z.Y. Liu, M.Q. Zhu, K.L. Zheng, *J. Mol. Struct. (Theochem)* 710 (2004) 175.
- [21] P. Politzer, J.S. Murray, in: J.S. Murray, K.D. Sen (Eds.), *Molecular Electrostatic Potentials: Concepts and Applications*, Elsevier, Amsterdam, 1996.
- [22] F.Y. Liu, B.L. Liu, Z.X. Meng, S.J. Zheng, *J. Mol. Struct. (Theochem)* 712 (2004) 207.